

TCS ASPIRE : UNIX quiz/question-answer 2012

Question 1 of 10 10.0 Points

"ls" command in UNIX

- A. Lists the files in a directory.
- B. Compare the size of two files
- C. Gives the currently logged session details.
- D. Compares two numerals and prints the lower value

Answer Key: A

Question 2 of 10 10.0 Points

Which among the following is not a part of UNIX operating system

- A. Programs
- B. Libraries
- C. Kernel
- D. Shell

Answer Key: B

Question 3 of 10 10.0 Points

In UNIX, the shell act as an interface between user and

- A. Kernel
- B. Operating System
- C. Hardware
- D. Programs

Answer Key: A

Question 4 of 10 10.0 Points

Everything in UNIX is

- A. None
- B. a Process
- C. a file or a process
- D. a File

Answer Key: C

Question 5 of 10 10.0 Points

Which command is used for changing directory in UNIX?

- A. mkdir
- B. cd
- C. chgdir
- D. cddir

Answer Key: B

Question 6 of 10 10.0 Points

In UNIX, file permissions are not set for

- A. User
- B. Other users
- C. Devices
- D. User Groups

Answer Key: C

Question 7 of 10 10.0 Points

In Unix, how can you read the manual – help pages – about a command

- A. Type 'help' and the command
- B. Type 'options' and the command
- C. Type 'details' and the command
- D. Type 'man' and the command

Answer Key: D

Question 8 of 10 10.0 Points

Which is the most common text editor in UNIX ?

- A. Visual editor
- B. notepad
- C. Word
- D. Open office

Answer Key: A

Question 9 of 10 10.0 Points

You have created a shell script and wants to run the script. What should you do before trying to run the script ?

- A. No action required
- B. Compile the script to create the executable
- C. Update the execute permission of the script
- D. Give ownership of the file to system admin

Answer Key: C

Question 10 of 10 10.0 Points

Which is valid mode of operation in vi editor ?

- A. Typing mode
- B. Command Mode
- C. Append mode
- D. Open mode

Answer Key: B

TCS ASPIRE : software engineering quiz/question answer 2012

Question 1 of 10 10.0 Points

Requirements can be classified in to two types, namely, functional requirements and non-functional requirements

- True
- False

Answer Key: True

Question 2 of 10 10.0 Points

Some examples of non-functional requirements are.

- 1.Number of significant digits to which accuracy should be maintained in all numerical calculations is 10
- 2.The response time of the system should always be less than 5 seconds
- 3.The software should be developed using C language on a UNIX based system
- 4.The matrix diagonalisation routine should zero out all off-diagonal elements, which are equal to or less than 10^{-3}
- 5.Experienced officers should be able to use all the system functions after a total training of two hours. After this training, the average number of errors made by experienced officers should not exceed two per day.

- True
- False

Answer Key: True

Question 3 of 10 10.0 Points

Every software system has the following essential characteristics:

- A. It has processes which collaborate with each other to generate the outputs.
- B. All the above
- C. It has a boundary. The boundary separates what is with in system scope and what is outside.
- D. The system may also use data stores to store data which has a life beyond the system.
- E. These processes operate on data by creating, modifying, destroying, and querying it.
- F. It takes inputs from external agents and generates outputs.

Answer Key: B

Question 4 of 10 10.0 Points

ERD complements DFD. While DFD focuses on data and the relationships between them, ERD focuses on processes and data flow between them

- True
- False

Answer Key: False

Question 5 of 10 10.0 Points

Polymorphism: – It is the means by which an operation behaves differently in different contexts.

- True
- False

Answer Key: True

Question 6 of 10 10.0 Points

Encapsulation: This helps in controlling the visibility of internal details of the objects. It improves security and integrity of data

- True
- False

Answer Key: True

Question 7 of 10 10.0 Points

There are 4 types of Abstraction, namely Entity, Action, Virtual machine & Coincidental.

- True
- False

Answer Key: True

Question 8 of 10 10.0 Points

There can be more than one hash fields on a file.

- True
- False

Answer Key: False

Question 9 of 10 10.0 Points

Concurrency refers to multiple transactions accessing the same database at the same time. In a system which allows concurrency, some kind of control mechanism has to be in place to ensure that concurrent transactions do not interfere with each other.

- True
- False

Answer Key: True

Question 10 of 10 10.0 Points

Software genetic development process contains three genetic phrases namely

- A. Software engineering, Definition, Coding
- B. Definition, development, maintenance
- C. Coding, design, Software engineering
- D. Design, Coding, Development
- E. Development, Definition, Testing

Answer Key: B

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TCS ASPIRE : software engineering quiz/question answer 2012

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Software generic development process contains three generic phrases namely

- A. Software engineering, Definition, Coding
- B. Definition, development, maintenance
- C. Coding, design, Software engineering
- D. Design, Coding, Development
- E. Development, Definition, Testing

Answer Key: B

TCS ASPIRE | Database management system quiz/question | latest 2012

Question 1 of 10 10.0 Points

Which of the following options list the steps to convert a table to its second normal form?

- A. All of the above.
- B. Find and remove fields that are related to the only part of the key.
- C. Assign the new table with the key i.e. part of the whole composite key.
- D. Group the removed items in the another table.

Answer Key: A

Question 2 of 10 10.0 Points

In which of the following ways does "TRUNCATE TABLE" differ from "DELETE"?

- A. All of the above.
- B. The number of deleted rows are not returned.
- C. Truncate operations are not transaction safe.
- D. Truncate operations drop and re-create the table which is much faster than deleting rows one by one.

Answer Key: A

Question 3 of 10 10.0 Points

Following tasks can be performed when using the ALTER TABLE clause:

- 1. Change the name of the table
 - 2. Change the name of the column
 - 3. Decrease the size of a column if table data exists.
- True
 - False

Answer Key: False

Question 4 of 10 10.0 Points

Sometimes tables within particular database become obsolete and ought to be discarded. Which of the following commands would you use for the same?

- A. None of the above.
- B. REMOVE TABLE < TABLE NAME>
- C. DROP TABLE < TABLE NAME >
- D. DELETE TABLE < TABLE NAME >

Answer Key: C

Question 5 of 10 10.0 Points

A foreign key must have a corresponding primary key or unique key value in the master table.

- True
- False

Answer Key: True

Question 6 of 10 10.0 Points

NULL value is equivalent to a value of Zero if the data type is number.

- True
- False

Answer Key: False

Question 7 of 10 10.0 Points

Integrity constraint can be dropped if the rule that it enforces is not longer TRUE or if the constraint is no longer needed.

- True
- False

Answer Key: True

Question 8 of 10 10.0 Points

SELECT TRIM(' Hansel ') "Trim both sides" FROM DUAL What would be the output of the above command?

- A. None of the above
- B. Trim both sides
- C. Hansel
- D. Trim both sides Hansel

Answer Key: D

Question 9 of 10 10.0 Points

What would be the output of following command. SELECT TRANSLATE('1sc1523','123', '7x9') "Change" FROM DUAL;

- A. 7sc1 5x9
- B. None of the above
- C. 7 sc1

D. Change 7 sct 5a9

Answer Key: D

Question 10 of 10 10.0 Points

What is joining a Table to itself called?

- A. Self Join
- B. Outer join
- C. Left inner join
- D. Inner join

Answer Key: A

TCS ASPIRE : Unix Assignment 1 quiz answers 2012

Question1:

Write a command to list all the files inside a folder i.e. if there is a folder inside a folder then it should list all files inside the sub-folder which is inside the folder to be listed.

Answer: ls -oR

[Hint]

Ls =list information about files.

-a =all (hidden files also), -R =recursive (list subdirectories)

Question 2:

Search all the files which contains a particular string, say "include" within a folder.

Answer: grep include ./*

[Hint]

Grep = searches the named input file for lines containing a match to the given pattern. ./ = in current directory (.), all files(*)

Question 3:

Rename all the files within a folder with suffix "Unix_" i.e. suppose a folder has two files a.txt and b.pdf then they both should be renamed from a single command to Unix_a.txt and Unix_b.pdf

Answer: for i in *.txt;do mv "\$i" Unix_"\$i".done;

[Hint]

Loop

Syntax for variable in no of times,

Do command;

Done;

Here: i= *.txt (for all files)

\$i to read value of variable

do mv "\$i" Unix_"\$i" = rename *.txt as Unix_*.txt

Question 4:

Rename all files within a folder with the first word of their content(remember all the files should be text files. For example if a.txt contains "Unix is an OS" in its first line then a.txt should be renamed to Unix.txt

Answer: for i in *.txt;do head -1 "\$i" > ad.txt ; x="\$(cut -d ' ' -f1 ad.txt)"; mv "\$i" "\$x.txt"; done;

[Hint]

i = *.txt (all text files);

do head -1 "\$i" > ad.txt, = prints first line and store in ad.txt file.

x="\$(cut -d ' ' -f1 ad.txt)" : = removes first column (field) delimited by space from ad.txt file stores in x.

mv "\$i" "\$x.txt" = rename *.txt as value of x .txt

Question 5:

Suppose you have a C project in a folder called "project". it contains .c and .h files, it also contains some other .txt files and .pdf files. Write a Linux command that will count the number of lines of your text files. That means total line count of every file. (remember you have to count the lines in .txt files only)

Answer: wc -l *.txt

[Hint]

Here: wc =word count

-l =to count lines

*.txt =all text files

Question 6:

Rename all files which contain the sub-string 'foo', replacing it with 'bar' within a given folder.

Answer: for i in \$(find /foo); do mv "\$i" "\${i/foobar}/"; done

Question 7:

Show the most commonly used commands from "history". [hint: remember the history command, use cut, and sort it.

Answer: history|sort|cut -d '^' -f1

TCS ASPIRE : Unix Assignment 2 quiz answers 2012

Question 1:

Create a tree structure named 'training' in which there are 3 subdirectories - 'level 1', 'level2' and 'cep'. Each one is again further divided into 3. The 'level 1' is divided into 'sdp', 're' and 'se'. From the subdirectory 'se' how can one reach the home directory in one step and also how to navigate to the subdirectory 'sdp' in one step? Give the commands, which do the above actions?

Answer : To navigate from 'se' to home directory in one step, the command is : `cd`

To navigate from 'se' to 'sdp' in one step, the command is : `cd ../b`

Question 2:

How will you copy a directory structure dir1 to dir2 ? (with all the subdirectories)

Answer : `cp -R dir1 dir2`

Question 3:

How can you find out if you have the permission to send a message?

Answer : `ls -l` (To check access permissions)

Question 4:

Find the space occupied (in Bytes) by the /home directory including all its subdirectories.

Answer : `du -s -b`

Question 5:

What is the command for printing the current time in 24-hour format?

Answer : `date +%T`

Question 6:

Create the following files: chapA, chapB, chapC, chapD, chapE, chap01, chap02, chap03, chap04, chap05, chap11, chap12, chap13, chap14, and chap15.

Answer :

The general syntax is :

cat > filename

Type the contents of file

Press Ctl+D

Example:

cat > chapA

Type the contents of file

Ctl+D

Question 7:

What is the command for printing the year, month, and date with a horizontal tab between the fields?

Answer : date +%Y%m%B%T%

Question 8:

With reference to question 7, What is the command for listing all files ending in small letters?

Answer : find -name "[a-z]."

Question 9:

With reference to question 7, What is the command for listing all files ending in capitals?

Answer : find -name "[A-Z]."

Question 9:

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Question 10:

With reference to question 7, What is the command for listing all files whose last but one character is 0?

Answer : find -name "*0?".